SR UNIVERSITY

School of Computer Science and Artificial Intelligence

ECO TRACK

“EcoTrack: A Smart Plastic Waste Recycling Tracking System”

A centralized digital platform to monitor and track plastic waste from collection to recycling, improving transparency, efficiency, and sustainable waste management.

B.Vyshnavi - 2403A510C8

G.Meghana – 2403A510C1

G.Siddhartha - 2403A510A8

B.Abhiram - 2403A510B0

SR UNIVERSITY

SOLUTION APPROACH

EcoTrack follows a data-driven and centralized approach to solve the problem of untracked plastic waste. The system focuses on capturing waste data at every stage, ensuring transparency, and providing meaningful insights for decision-making.

The approach is built on three key ideas:

- Data Collection: Accurate entry of plastic waste details by collectors
- Real-time Tracking: Continuous update of waste status from collection to recycling
- Data Visualization: Clear representation of data through dashboards, charts, and heatmaps

By integrating these components, the system improves coordination between stakeholders and enables efficient waste management.

Additionally, the solution emphasizes:

- Simplicity: Easy-to-use interface for quick data entry
- Scalability: Can be extended to smart city systems
- Impact Analysis: Helps measure environmental benefits like recycling efficiency

SR UNIVERSITY

DESIGN EXPLANATION

Home Section

Acts as the entry point of the application.
Provides navigation and a quick overview of system activities.

Collector Page

Allows collectors to input waste data such as area, quantity, and date.
This is the starting point of the data flow.

Recycling Center Page

Used to update the status of collected waste.
Marks waste as Recycled, completing the tracking cycle.

Dashboard

The core module that displays:

- Total collected waste
- Total recycled waste
- Pending waste
- Efficiency percentage

Includes charts for better understanding of data.

SR UNIVERSITY

Heatmap

Provides a visual representation of waste distribution.
Helps identify high-waste areas quickly, enabling better planning.

Reports Section

Generates summaries of system data.
Helps analyze trends and evaluate recycling performance.

Complaint Section

Allows users to report waste-related issues.
Improves system responsiveness and accountability.

System Flow

Collector → Data Storage → Recycling Update → Dashboard / Heatmap / Reports